Daily Drilling Report
SPD19B-03
4/14/2014

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
20.0	16"	41.00	72.83	31.70	105.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
18 5/8" Shoe, 870.00mKB	221,095.38	2,103.42	4,671,550.25		17.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
1,478.00	1,492.00		14.00	6.68	2.1

Operations at Report Time
R/U 13 3/8" Casing handling tools.

Operations Summary
RIH directional BHA#6 from surface to bottom (1478m). Resume drilling 16" hole to 1492m (Section TD). Sweep hole. Spot Hi-Vis pill. POOH to 750m.

Operations next 24hrs
Cont' POOH directional BHA#6 to surface. Run and Land 13 3/8" Casing.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Mohammad Ali Rezaei	Senior Drilling Sup.
Vahab Kari	Night Drilling Sup.
Hamed Daneshkhah	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	3/9/2014	4/29/2014
BOP Test	4/7/2014	4/29/2014
Chemical Spill Drill	3/29/2014	5/29/2014
Fire Drill	4/12/2014	6/2/2014
H2S Drill	4/4/2014	5/5/2014
Heli Crash Drill	4/14/2014	6/14/2014
Medical Drill	4/4/2014	6/4/2014
MOB Drill	2/28/2014	4/30/2014
Pit Drill	4/9/2014	5/25/2014
Trip Drill	4/8/2014	5/29/2014

STOP Cards

Number of Reports: 34

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	01:30	1.50	16"	D	24	N	(NPT for HPOGC Company) L/D used 9 5/8" motor. P/U new 9 5/8" motor, B/O and L/D Slick Protector. M/U new 15 3/4" Sleeve stabilizer.
01:30	02:30	1.00	16"	D	04	P	RIH directional BHA#6 consists of: New 16" Insert bit (CR14JMRS, 435, SN: 1244888, Nozzles: 3x20 + 1x16) + New 9 5/8" Motor (BH: 1.25 , Sleeve stabilizer: 15 3/4") + 14 3/4" String stabilizer + 9 1/2" Restrictor + 9 1/2" NMDC + 9 1/2" TM - Sub , to 24m. Align motor with MWD.
02:30	03:00	0.50	16"	D	24	N	(NPT for HPOGC Company) Connect TDS. Shallow test new mud Motor (730 gpm, 850 psi) , OK.
03:00	05:00	2.00	16"	D	04	P	Cont' RIH rest of BHA consists of: 1 jt 9 1/2" NMDC + 9 1/2" Float sub + XOS + 8 jts 8 1/4" DC + 8" Jar + 6 jts 8 1/4" DC + XOS + 15 jts 5 7/8" HWDP , to 313m.
05:00	09:30	4.50	16"	D	04	P	RIH directional BHA#6 on 5 7/8" DP f/ 313m to 1445m. W/R down w/ 300 gpm/ 20 rpm and tag bottom (1478m).
09:30	16:00	6.50	16"	D	03	P	Profile new Bit with low parameters. Resume directional drilling 16" hole section in rotation mode f/ 1478m to 1492m w/ 800 gpm/ 1800 psi/ 35-45 kips/ 50 rpm/ 8-15 kft-lbf.
16:00	18:30	2.50	16"	D	07	P	Sweep hole by pumping 30 bbl Lo-Vis + 30 bbl Hi-Vis, circulate 30 min, pump 35 bbl Lo-Vis + 35 bbl Hi-Vis w/ 900 gpm/ 2050 psi. Circulate till shakers clean, meanwhile reciprocate string with 35 rpm.
18:30	19:00	0.50	16"	D	09	P	Spot 150 bbl Hi-Vis pill on bottom w/ 630 gpm/ 1100 psi.
19:00	00:00	5.00	16"	D	04	P	POOH directional BHA#6 f/ 1492m to 750m.
00:00	00:00						Apr 15 , 2014 00:00 - 04:30 Cont' POOH directional BHA#6 f/ 750m to surface. L/D 8" Jar. R/B BHA. Check the bit (1-1-WT-G-E-I-NO-TD). 04:30 - 05:00 P/U & M/U Aker Wear Bushing R/T to one std of 5 7/8" DP, RIH to 19.45m, retrieve WB, pull string to surface, R/B std. 05:00 - 05:30 P/U & M/U & RIH Aker washing tool and wash inside BOP & WH w/ 5 rpm/ 550 gpm/ 170 psi. POOH to surface and L/D same. 05:30 - 06:00 L/D drilling Bail. M/U 5 ft pup joint + XOS + AL-Ghaith Fill up tools to TDS.

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Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
1,492.00	Polymer Mud	9.20	43	12.0	20.000	8.8
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
8.8						
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
5	6	0.0	92.0	8.0	24,200.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
5.000	6.000		0/100	49.0	1759.0	1533.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
2248.0	72.0	100.0			13.09539	0.409

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.50	6.	140	4.0	1,005.0	35.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Mataf Victory	Crew Boat	4/14/2014 21:45	4/14/2014 22:40	Delivered cargo and embark 1 Pax for Kish, Dep to Aban 8.
Ahmed-6	Supply Boat	4/14/2014 12:00	4/14/2014 15:30	Delivered all DD6 Cargo & FW-50 MT, C/Off & Dep to Sahar 2.
Navis King	Supply Boat	4/14/2014 01:00		Stdby boat.

Mud Additives

Description	Consumed	Cum Rec	On Loc
AMYLOSE B		1,000.0	563.0
CLAY-BOND L		104.0	82.0
Bio-Cide	2.0	274.0	101.0
Defoam SLC	2.0	220.0	62.0
Bio-XCD	8.0	454.0	126.0
Caustic Soda	10.0	825.0	226.0
Pac-R		440.0	181.0
MICA M		160.0	130.0
MICA F		160.0	160.0
Pac-UL		310.0	158.0
Lime	11.0	338.0	117.0
WALNUT F		160.0	160.0
WALNUT M		200.0	170.0
Soda Ash	3.0	312.0	98.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Barite BB	0.0	0.0	15.0
Barite Bulk	0.0	0.0	147.0
Cement G	0.0	0.0	168.0
Bentonite Bulk	0.0	0.0	65.0
Bentonite BB	0.0	0.0	21.0
Diesel Fuel	12.0	0.0	110.0
Potable Water	20.0	20.0	110.0
Drill Water	35.0	50.0	428.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
6	16	Varel	CR14JMRS	1244888	Insert	435	20/20/20/16
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
1.11	6.68	61,723.20	1,478.00	1,492.00	14.00	2.1	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
800	800	1,800.0	1,800.0	42	42	154	154
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
9.96	6.68	100.00	0.00	349	311	269	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
1,478.00	1,492.00	6.68	15,000.0	800	50	1,800.0	42	311

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
15 jts 5 7/8" HWDP	5 7/8	4.250	139.60	313.30
XOS	8	3.000	0.79	173.70
6 jts 8 1/4" DC	8 1/4	2.810	56.24	172.91
8" Drilling Jar	8	3.500	5.72	116.67
8 jts 8 1/4" DC	8 1/4	2.810	75.34	110.95
XOS	9 1/2	3.000	1.22	35.61
9 1/2" Float Sub	9 1/2	3.000	0.78	34.39
9 1/2" NMDC	9 1/2	4.000	9.44	33.61
9 1/2" TM - Sub	9 1/2		2.11	24.17
9 1/2" NMDC	9 1/2	4.000	9.44	22.06
9 1/2" Restrictor sub	9 1/2	4.310	0.92	12.62
14 3/4" Stabilizer	14 3/4	4.000	2.16	11.70
9 5/8" Motor (BH: 1.25, Sleeve: 15 3/4")	9 5/8		9.14	9.54
16" Insert Bit	16		0.40	0.40

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Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0

Applied Surface Pressure (psi)	Volume Pumped (bbl)	Current MAASP	Kick Tolerance	Dens Fluid (lb/gal)	Leak Off Pressure (psi)
400.0	8.0	400.00		11.48	1,650.4

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
1,472.50	247.70	16.30	1,405.74	-176.02	-348.07	390.05	0.98	-0.83

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
Upper Gadvan		1,209.00	1,156.52
Khalij		1,231.00	1,176.98
Lower Gadvan		1,247.00	1,191.88
Fahliyan		1,286.50	1,228.87
Hith		1,469.30	1,402.67

General Remarks

Comment

>>> Requirements: (20 3/4" Dovetail for BOP Test Plug)

>>> Sent:

*** Mataf Victory boat to Kish base: (HPOGC D&M tools: 1 no 9 5/8" Motor + 1 no 15 3/4" Sleeve stabilizer + 1 no 15 7/8" Sleeve stabilizer + 1 no Slick protector) + (AL-Ghaith 2 no Spider slips).

>>> Received:

*** Mataf Victory boat f/ Kish base: (HPOGC D&M tools: 2 jts 9 1/2" NMDC + 1 no 9 1/2" Transmitter sub + 1 no 9 1/2" Restrictor sub + 1 no 9 1/2" Float sub + 1 no 15 7/8" Sleeve stabilizer) + (AL-Ghaith tools: 1 no FMS + Control Panel)

*** Ahmed-6 boat f/ Kish base: (PDF Chemicals: 5 pallet Starch + 2 pallet Pac-LV + 2 pallet Pac-R + 2 pallet Clay-Bond + 2 pallet Mica-F + 2 pallet Mica-M + 2 pallet nut Plug-F + 3 pallet Nut Plug-M) + (3 no used 16" PDC Bit) + (12 no Carrier assy, 2" nominal) + (28 no Shaker Screen, API-230) + (41 no Shaker Screen, API-170) + (29 no Shaker Screen, API-200) + (HPOGC 9 5/8" Motor to be B/O from 16" Bit) + (50 MT DW).